Learning Surrogates to Plan Paths for Information Gathering about Hazards in Dangerous Environments

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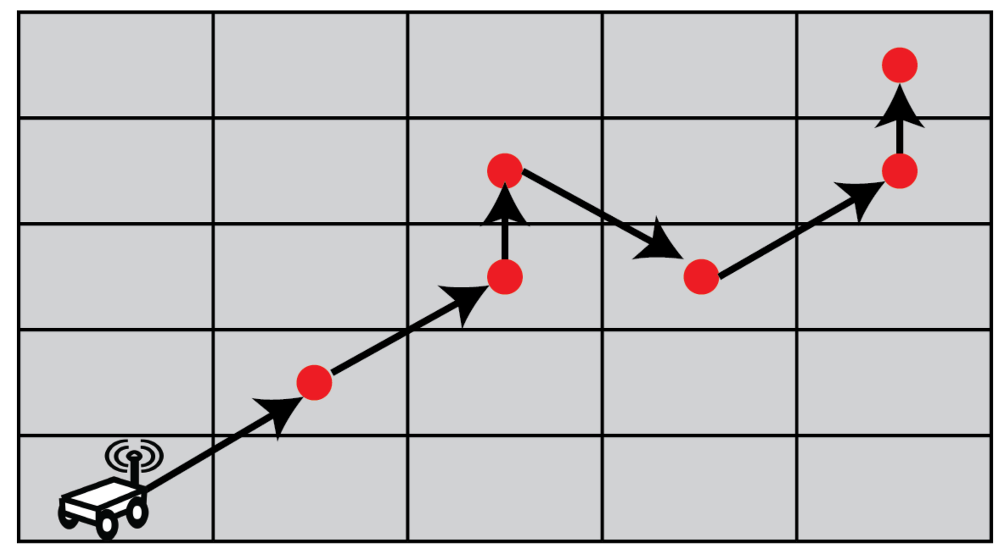
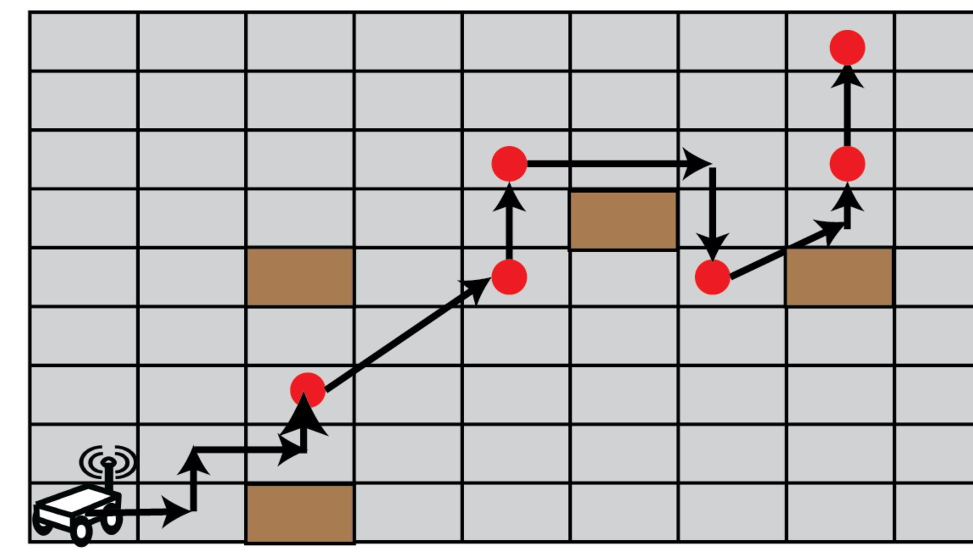
What? and Why?

- Information-gathering robots need to replan their missions upon detecting lethal hazards to increase their survival chances.
- Computational complexity and submodular nature of Shannon information theoretic planners prohibits quick replanning.
- We propose a **machine learning-based approach** to ensure real-time re-planning.

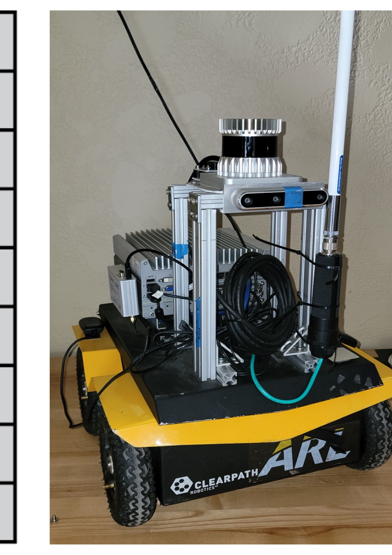
Planner's Approach To Gather Information and Navigate



(a) Proposed methodolgy generates informative waypoints for the lower level planner



(b) Lower-level planner uses D* lite to navigate waypoints and avoid dynamic obstacles

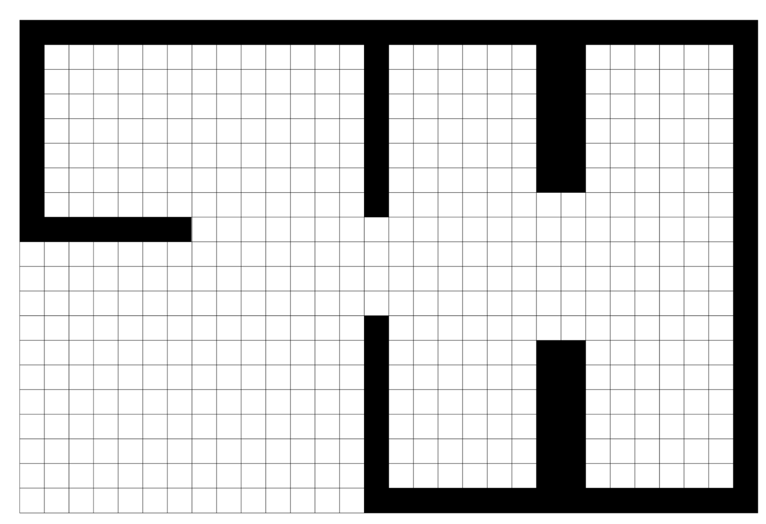


Jackal ground robot

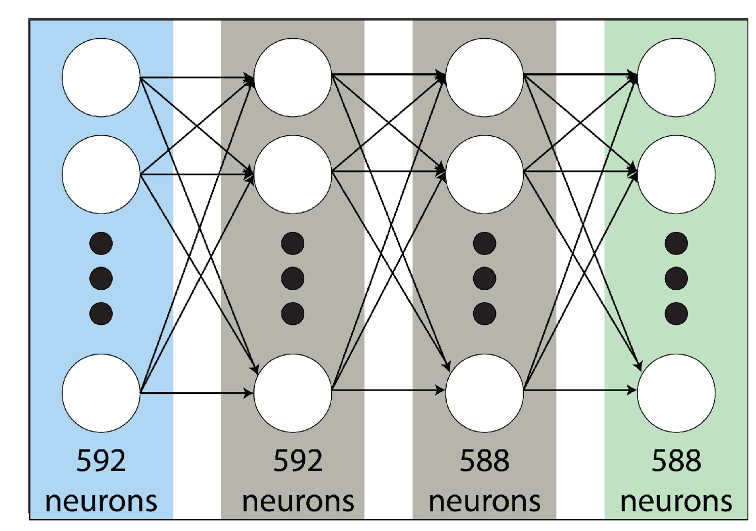
Our Contributions

- We train an ML surrogate model based on data from Shannon information theoretic planner.
- The **neural network model** provides an efficient approximation to the theoretical method.
- This reduces the computational cost of **real-time information gathering**.
- Enables faster online reaction to new information about hazards – increasing robot's survival rate

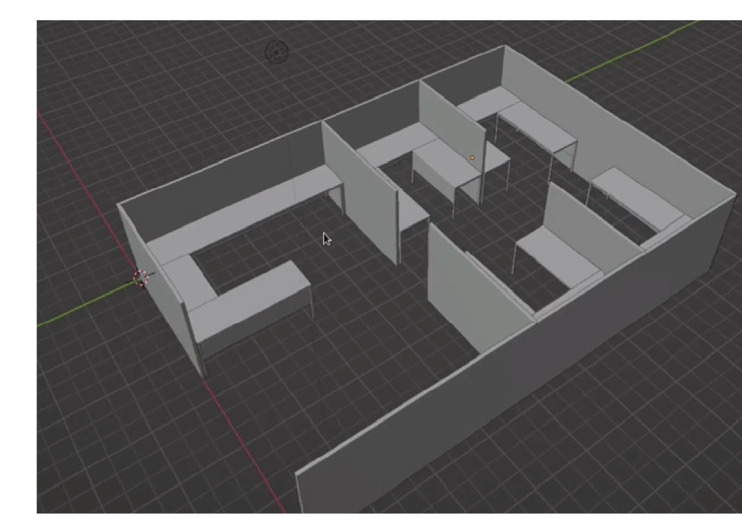
Simulation Environment used for generating Training dataset



(a) Discretized environment used for informative planning and generating informative waypoints



(b) Neural network architecture used by Machine Learning surrogate for generating informative waypoints



(c) Simulated office space environment used in Army Research Lab's Pheonix Stack Unity Simulator

Methodology

- **Two-step grid-based planner**:
  - High-level planner generates information-gathering waypoints
  - Low-level planner navigates to the waypoints while avoiding dynamic obstacles
- **Information theoretic planner (ITP):**
  - **Notation:** Search space $\mathbf{S}$, Cells C, Hazard belief map Z, Observations of robot-destruction $\boldsymbol{\Delta}$
  - Robot's path ζ at deployment $-d$: $\zeta_d = \langle C_1, C_2, \ldots, C_l \rangle \subset \mathbf{S}$, $\zeta_d(1) = C_{\text{start}}$ and $\zeta_d(l) = C_{\text{goal}}$; l – fixed number of grid-steps
  - The planner's task is to calculate a path $\zeta^\star$ to **maximize information** about the presence of hazards Z in search space $\mathbf{S}$

$$\zeta_d^\star = \arg\max_{\zeta_d} I(Z; \boldsymbol{\Delta_d})$$

where $I(Z; \boldsymbol{\Delta_d})$ denotes the **mutual information** of the belief map Z and the robot destruction observation $\boldsymbol{\Delta_d}$.

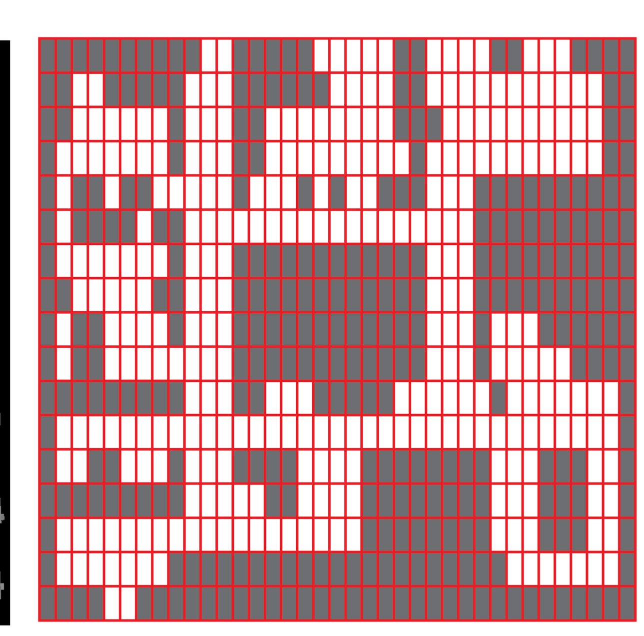
- **ML surrogate model:**
  - **Multi-layer perceptron** with sigmoid activation function
  - Outputs a score correlated to the **likelihood** that a given cell is along the optimal search path

Hardware tests in an Indoor Environment

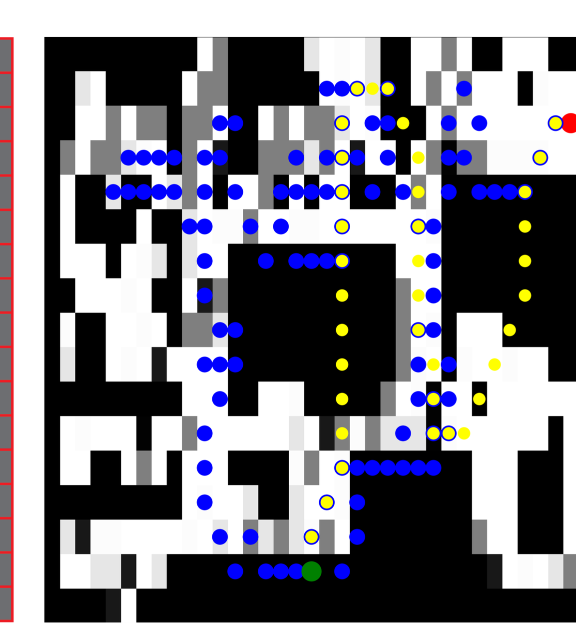
Clearpath Robotics Jackal navigating in ARL's office space

Search Grids of an Environment from Real-world experiments

LiDAR Map



Discretization

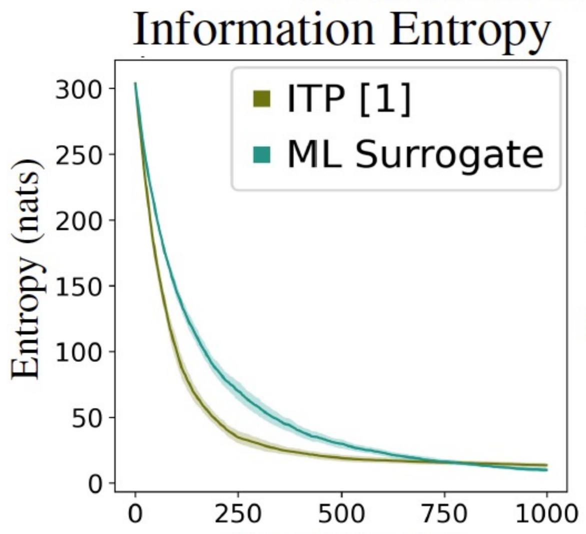


Waypoints

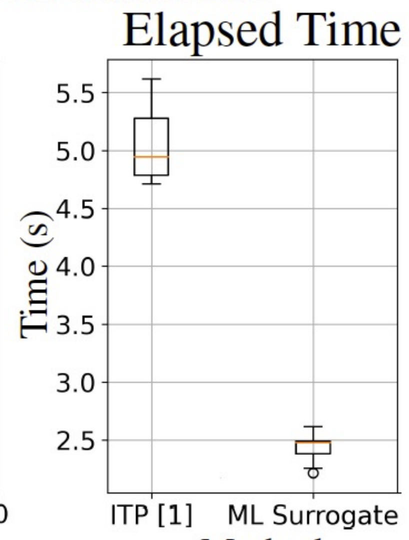
Quantitative Results

Simulation Environment

Information Entropy

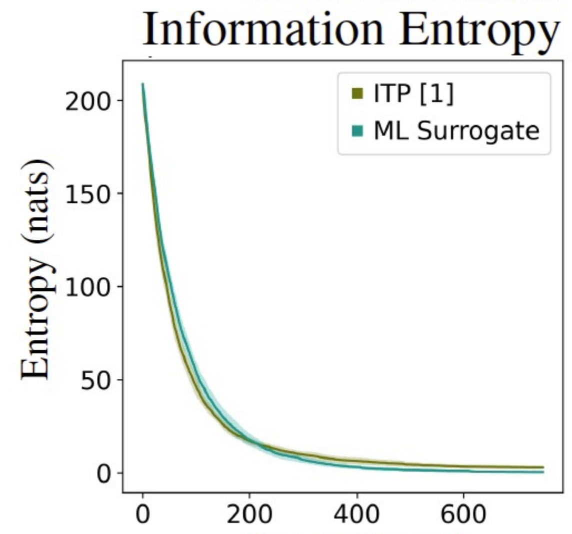


Elapsed Time

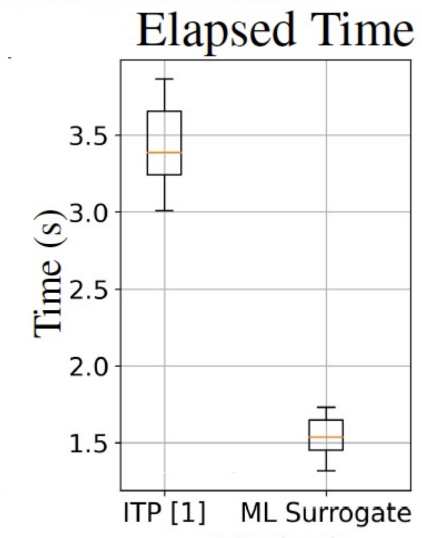


Demonstration Environment 1

Information Entropy



Elapsed Time



20 Monte Carlo trials

Insights: In simulations and hardware tests, both methods show **similar convergence** rate, but ML surrogate model offers **two-fold decrease** in computation time.

Experimental Verification

- **Performance Metrics:**
  1. Change in Shannon information entropy
  2. Total computation time to gather information about hazards
- Robot's **observation uncertainties** is modelled through false positives and false negatives.

Key Takeaways

- An effective **ML-based surrogate** alternative to **Shannon** information-theoretic planner
- **D*Lite** path planner to navigate in dynamic workspaces

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